Lamin Jatta

AI Software Engineer & Researcher

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I develop AI software for technical documents and engineering tasks. At Novia, I work on Python services, agent tools, simulation experiments, and React interfaces. My data-science work at JUICER included forecasting and price recommendations. I hold two master's degrees in Computer Science.

Technical skills

AI and ML: Python, R, PydanticAI, embeddings, retrieval-augmented generation (RAG), multimodal extraction, TensorFlow, PyTorch.

Software and delivery: FastAPI, Pydantic, MCP, React, TypeScript, Git, Docker, GitHub Actions, pytest, Ruff.

Data and cloud: SQL; Azure Databricks, Azure OpenAI, website hosting, and VM monitoring; GCP, Vertex AI, BigQuery, Cloud Run; MLflow, DVC.

Professional experience

Project Researcher

Novia University of Applied Sciences | Feb 2024 - present | Turku, Finland

Maritime Technology RDI and the Virtual Sea Trial project.

- Co-developed ANCHOR with Christoffer Björkskog and the Novia team, using Python/FastAPI and React/TypeScript. I worked on source-linked document extraction, runtime integration, extension discovery, and synchronization.
- Implemented table-caption matching based on page layout and JSON Schema validation before MCP tool calls. I also worked on cell-level source links and PDF highlighting so users could check extracted values.
- Implemented the initial PydanticAI tool integration for Agent-in-the-Loop and contributed to experiment execution, reporting, and evaluator fixes. The team's six-task benchmark used Functional Mock-up Unit (FMU) simulations, deterministic checks, and GPT-5-mini on Azure OpenAI.
- Contributed to layout evaluation, common extraction paths, and source highlighting in MERI's earlier research code. Co-authored its peer-reviewed paper on extracting parameters from engineering documents.

Data Scientist

JUICER | Aug 2022 - Aug 2024 | United States

Restaurant pricing and margin-management software.

- Collected and prepared data, performed exploratory analysis, and built visual reports for clients.
- Engineered features and deployed machine-learning models for time-series forecasting and price recommendations.

AI Software Developer Intern

SUPPLYZ Oy (now RESONIKS) | Dec 2022 - May 2023 | Tampere Metropolitan Area, Finland

Acoustic inspection and non-destructive testing using AI.

- Analyzed acoustic and relational datasets and developed machine-learning models using TensorFlow and PyTorch.
- Worked with MLflow, DVC, Vertex AI, BigQuery, and Cloud Run during model development and deployment on GCP.

Earlier experience

Assistant System Administrator

Guaranty Trust Bank (Gambia) Ltd | Oct 2017 - Oct 2019 | Fajara, The Gambia

Maintained Active Directory and server applications, tested and debugged banking systems, ran end-of-day backups, and supported application users.

Web Developer

Xanda Limited | Jan 2017 - Oct 2017 | London, United Kingdom

Contributed to WordPress website development, implementation, testing, and release within the development team.

Quality Assurance Technician

International Consumer Services Sweden AB / alaTest | Apr 2016 - Sep 2016 | Stockholm, Sweden

Checked review-to-product matching, provided feedback to production teams, and worked on delivery-flow and pipeline problems.

Lecturer and Course Trainer

Management Development Institute | Jan 2014 - Nov 2015 | The Gambia

Delivered Cisco networking training covering network cabling and inter-networking for homes and small to medium-sized organizations.

Education

Master's degree in Computer Science

Åbo Akademi University | 2024

Authored a thesis on maritime speech-transcript post-processing using sentence embeddings, Standard Maritime Communication Phrases, and a fine-tuned language model. Thesis.

Master's degree in Computer Science

Ca' Foscari University of Venice | 2022/2023

Thesis: A comparison of classification algorithms. Compared Naive Bayes, k-nearest neighbours, logistic regression, and linear/quadratic discriminant analysis using simulations and a real dataset in R. Thesis record.

Selected publications and accepted work

Agent-in-the-Loop: Using AI Agents to Perform Control-Oriented Simulation Tasks

Co-author with Christoffer Björkskog and Mikael Manngård. European Control Conference (ECC), 2026, pp. 3109-3115.

Towards Automated Parameter Extraction from Engineering Documents

Co-author with Christian Möller, Christoffer Björkskog, Andreas Lundell, Mikael Manngård, and Johan Westö. IFAC-PapersOnLine 59(24), 197-202, 2025.

Agentic Generation of GUI End-to-End Tests for Microservice-Based Systems

Second author. Accepted full paper, QUATIC 2026. I contributed structured specifications, execution briefs, browse traces, and evaluation reporting to the collaborative software project.

Code and engineering case studies

ANCHOR | MERI | Agent-in-the-Loop

Languages: English (full professional proficiency); Finnish, Swedish, French, and Italian (A2, basic communication).